Smart Construction Platform Proposal

An Intelligent Unified Environment for Construction Project Management, Engineering Coordination, and Stakeholder Collaboration

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Project Overview

The Smart Construction Platform is a unified project management and collaboration platform for construction projects. Its main purpose is to connect everyone involved in a project in one shared environment, including:

- Owner or client representative
- Contractor and subcontractors
- Engineers
- Consultants
- Project manager
- Site engineers
- Office administrator
- Reviewing engineers and technical directors

Instead of managing project information through separate tools, chats, phone calls, files, and social media groups, the platform brings project tasks, schedules, documents, drawings, field reports, design changes, approvals, communication, budget information, engineering models, and project evidence together in one connected system.

The platform is based on roles and permissions. The administrator can define roles and access according to the organization's structure. Each participant sees and works on what is relevant to their responsibility.

The platform also introduces AI-assisted workflows to reduce repetitive work and help users find and process project information faster. Voice input can be used for field reporting. AI can search project documents and provide specialized assistance. However, AI does not replace professional judgment. Important changes require permission checks and human confirmation.

Engineering model information is part of the platform. The current scope focuses on IFC-based models. IFC models can be viewed and compared across revisions, allowing relevant design changes to be connected with tasks, issues, milestones, and field activities. DWG support is not included in the current implementation scope.

Problem Statement

Project Problems

- Manual daily handling of large project data.
- Scattered project information across tools, files, conversations, and social media.
- Difficult search, review, documentation, and auditing.
- Disconnected project information.
- Cross-discipline impact of design changes.
- Time-consuming field reporting.
- Late detection of delays and schedule risks.
- Fragmented communication among stakeholders.

- Limited owner visibility into budget and spending.
- Hard-to-reconstruct decisions, approvals, and updateatas.
- Many distributed data types: issues, designs, reports, contracts, photos, audio, models.
- Current engineering model scope is IFC; DWG is not supported.

Data Related Problems

- No unified data layer connecting issues, photos, tasks, documents, approvals, and evidence.
- Weak relationships between project records.
- Fragmented evidence for issues, changes, and decisions.
- Duplicate and inconsistent records across systems.
- Inconsistent permission checks across linked data.
- Incomplete audit trail across connected records.
- Difficult retrieval of all records related to one issue, task, or change.
- Limited connected views for reports, AI, and notifications.
- Poor integration between field data and office data.
- No single source of truth for connected project items.

Proposed Solution

The proposed solution is a single connected platform where project participants work with the same project information while receiving access according to their roles and permissions. The platform connects office management, engineering coordination, construction activities, and field work instead of treating them as separate systems.

A typical project flow is designed to work as follows:

- The administrator creates the project structure, users, roles, and permissions.
- The project manager organizes members, tasks, milestones, schedules, documents, budget information, and project activities.
- Engineers, site engineers, contractors, consultants, and other participants work on the parts of the project assigned to them.
- Project information such as documents, drawings, photos, audio, issues, design changes, and engineering models is stored within the project library.
- When a design or model revision is introduced, the platform can compare it with an earlier version and identify relevant changes.
- Potentially affected tasks, issues, or milestones can be proposed for review by the responsible project member.
- Field users can submit photographs and voice reports, which can be processed into useful project information.
- AI can assist with transcription, document questions, analysis, and selected project actions, while important actions require user confirmation.

- Approved changes, updates, expenses, reports, and decisions remain connected to their supporting evidence and project history.
- Relevant project participants receive web or mobile notifications when important events require

Project Objectives

- Create one connected project environment for owners, contractors, engineers, consultants, project managers, administrators, and other project participants.
- Centralize project tasks, schedules, documents, drawings, field observations, issues, design changes, approvals, budget information, and evidence.
- Provide a flexible role-based permission system that allows organizations to define roles and access according to their own structure.
- Support task dependencies, milestones, critical-path analysis, progress tracking, and schedule-change recording to make project delays easier to identify and manage.
- Support field reporting through photographs and voice input to reduce the effort required to record site activities.
- Use AI to transcribe and interpret voice input, answer questions about authorized project documents, and assist with selected project-analysis tasks.
- Support AI-assisted structured actions while keeping users in control through permission checks and confirmation before important changes are applied.
- Support IFC-based engineering models, model viewing, revision comparison, and connections between model changes and project work.
- Provide project budget and spending visibility so authorized owners and project participants can track spending, remaining budget, and supporting information.
- Maintain a traceable history of important project activities, decisions, updates, approvals, and evidence.
- Provide Arabic and English interfaces, including right-to-left support, and design the voice workflow.

Users and Stakeholders

The platform is designed for projects that involve several internal and external participants. Access is controlled according to each participant's role and project responsibilities.

Stakeholder	Type	Main Responsibilities
Office Administrator	Internal	Manages users, roles, permissions, disciplines, and organization-level access.
Project Manager	Internal	Manages project members, tasks, schedules, milestones, budget visibility, reviews, and major project decisions.
Engineer	Internal	Works on assigned tasks, updates progress, raises issues, uploads information, and performs permitted reviews.
Site Engineer	Internal	Records site activities, field evidence, observations, photos, and voice reports.
Reviewing Engineer / Technical Director	Internal	Reviews tasks, reports, design changes, evidence, budget-related submissions, and selected AI findings.
Owner / Client Representative	External	Views authorized project information, monitors progress and spending, reviews evidence, and raises requests.
Contractor / Subcontractor Representative	External	Updates assigned work, submits field evidence, raises issues, and accesses shared project information.
Consultant	Internal	Reviews and responds to project information, technical submissions, and approvals according to assigned permissions.

Expected Platform Features

The proposed platform is expected to include the following major features:

1. **Unified Project Management**

Project creation and management with project members, parties, milestones, settings, and project information.

Task creation, assignment, priorities, progress, planned and actual dates, comments, reviews, and status tracking.

Task dependencies and critical-path analysis to help identify activities that can affect the project schedule.

2. **Role-Based and Configurable Permissions**

Role-based access control (RBAC) for internal and external users.

Administrator-controlled roles, permissions, disciplines, and project access.

Project-specific access and user-level overrides when required.

A flexible permission model that can adapt to different organizations and project structures.

3. **Central Project Library**

A centralized library for drawings, reports, contracts, specifications, invoices, BOQ files, schedules, technical documents, photos, audio recordings, engineering models, and other project evidence.

Permission-controlled access to project files and information.

File classification and content-based organization to make project information easier to find and manage.

4. **Field Reporting and Evidence**

Site reports and field submissions for recording activities and observations.

Photographs and other evidence connected to project activities.

Voice-based reporting through the mobile workflow.

Permission-controlled verification and review of submitted field evidence.

5. **Voice and AI Assistance**

Voice transcription for field reports and other supported project inputs.

Interpretation of natural spoken information in Arabic, English, and mixed-language communication.

Support for naturally spoken local Arabic, including Palestinian conversational speech, as a target of the voice workflow.

AI-assisted structured actions such as suggesting a progress update, issue, observation, or other supported action.

User confirmation before important AI-suggested changes are applied to project data.

6. AI Document Assistant and RAG

Question answering over project documents using Retrieval-Augmented Generation (RAG).

Answers based on information retrieved from the project's authorized document library.

Permission checks before retrieval so AI cannot expose information that the user is not authorized to access.

Source-connected AI responses and reviewable findings.

7. Specialized AI Agents

Site Report Agent to assist with field-report information.

Document Consistency Agent to identify possible inconsistencies across project documents.

Revision Impact Agent to help analyze the possible project impact of design or model changes.

Request / RFI Agent to assist with project requests and related information.

Schedule Risk Agent to help identify potential schedule risks.

Agents are intended to reduce repetitive work and highlight information that may require attention, not replace professional judgment.

8. IFC / Engineering Model Intelligence

Upload and management of IFC-based engineering models.

Model information extraction and organization.

Comparison between model revisions to identify added, removed, modified, or moved elements where supported by the available model evidence.

Connection of important model changes to tasks, issues, milestones, and project activities.

Coordination and data-quality findings presented as information to be reviewed rather than automatic engineering decisions.

9. Design Change and Review Workflows

Design changes can move through proposal, review, approval or rejection, and implementation.

Issues can be recorded with status and severity.

Consultants and reviewing engineers can participate in permission-controlled review workflows.

Important decisions and supporting evidence remain part of the project history.

10. Budget and Spending Management

Authorized users can define and monitor a project budget.

Project spending can be recorded against the available budget.

Owners can view how much has been spent, how much remains, and where recorded spending originated.

Expenses can be connected to supporting information or evidence to improve transparency and traceability.

11. Communication and Notifications

Direct, group, project, and contextual communication.

Web application notifications and mobile push notifications.

Automatic reminders for selected project events and deadlines.

Notifications for relevant assignments, approvals, issues, design changes, schedule updates, and other important events.

12. Audit Trail and Project Traceability

Recording of important project activities and changes.

Traceability of relevant actions, users, times, decisions, and supporting evidence.

Activity history that helps project participants reconstruct what happened during the project.

13. Security and Access Protection

Authentication and controlled access to project information.

Protection of sensitive actions through appropriate verification mechanisms.

Rate limiting, token control, audit logging, and permission checks across major platform workflows.

14. Arabic and English Support

Arabic and English user interfaces.

Right-to-left support for Arabic.

Bilingual project communication and information workflows where applicable.

How the Features Work Together

The main idea of the platform is not simply to provide many features, but to connect them through one project workflow. A change or action in one part of the project can therefore become relevant to other parts of the project.

Example: Design Change to Project Action

1. A new engineering model or drawing revision is uploaded to the project library.
2. The platform compares the new revision with the previous version and identifies relevant changes.
3. Potentially affected tasks, issues, or milestones can be proposed for review.
4. The project manager or responsible engineer reviews the proposed impact and decides what action is required.
5. A site engineer can later record the actual site situation using photos or a voice report.
6. The voice input can be transcribed and interpreted by the AI workflow, which may suggest a structured update.
7. The responsible user reviews and confirms the suggested action.
8. The project information is updated and the relevant users are notified.
9. The related actions, decisions, and evidence remain available in the project history.

AI Workflow

AI is proposed as an assistant within the platform rather than an independent decision-maker. Its purpose is to reduce repetitive work, improve access to project information, and help users identify information that may require attention.

- **Voice-to-Action Workflow**
  1. A user records a voice message, especially during field work.
  2. The audio is transcribed into text.
  3. The AI interprets the content and identifies relevant project information.
  4. The system maps the interpretation to a supported structured action, such as a progress update, field observation, or issue.
  5. The proposed action is checked against the user's permissions.
  6. The user reviews and confirms the action before the project data is changed.
  7. The resulting action and relevant source information can remain recorded as part of the project history.
- **Document Question Answering and RAG**

  The proposed RAG workflow will allow authorized users to ask questions about documents stored in the project library. The system retrieves relevant document content and uses it to generate an answer. Permission checks are applied before retrieval, so the AI workflow follows the same access boundaries as the rest of the platform.
- **AI Agents**

  Specialized agents will focus on bounded project activities rather than attempting to manage the entire project autonomously. Their role is to reduce repetitive analysis, organize information, identify possible risks or inconsistencies, and provide useful findings for human review. Where an AI result is uncertain, the system is intended to present it as a finding or suggestion rather than as a confirmed professional conclusion.

Budget, Spending, and Transparency

The proposed platform includes budget visibility as part of project management. For example, if an owner assigns a project budget of 20,000 NIS, recorded expenses can reduce the available balance and allow the owner to see the amount spent and the remaining amount.

The platform is intended to provide more than a single total. Each recorded expense can be associated with its source and supporting project information, allowing authorized users to understand where the money was spent and maintain a traceable record of project spending.

System Architecture and Technology

The proposed platform is designed as a layered web and mobile system. The web and mobile clients communicate with a FastAPI backend, which contains the main business rules, authorization, workflows, and integrations. PostgreSQL is used as the main project data store.

Part	Technology	Purpose
Backend	Python, FastAPI, Pydantic, SQLAlchemy, Alembic	API, business rules, permissions, workflows, and integrations.
Database	PostgreSQL	Project, user, task, document, review, issue, budget, and audit information.
Web	React, TypeScript, Vite, Tailwind CSS	Office, management, administration, documents, schedules, communication, and project workspace.
Mobile	Flutter / Dart	Field reporting, voice capture, photographs, tasks, notifications, and selected project information.
Engineering Models	IfcOpenShell, Three.js	IFC processing, model information, geometry handling, and 3D viewing.
AI	OpenAI SDK	Voice transcription, structured interpretation, embeddings/RAG, document question answering, and AI-assisted analysis.
Delivery	Docker Compose	Development and deployment setup for the main services.

Main Project Workflows

1. **Task Execution and Review**

A task is created, scheduled, assigned, and progressed through its workflow. Dependencies and schedule information help identify relationships between activities. When review is required, the task passes through the appropriate review process before completion.

2. **Field Evidence**

A field user submits photographs and, when needed, a voice recording. The information can be reviewed or verified by an authorized user and connected to the relevant project activity.

3. **Model Revision**

A new IFC model version is uploaded and compared with an earlier version. Relevant changes can be recorded and proposed as impacts on tasks, issues, or milestones. The responsible project participant decides whether a proposed impact should become actual project work.

4. **Document Questions**

Project documents are organized in the project library. Authorized users can ask questions about available documents, and the AI retrieves relevant information before generating a response.

5. **Budget and Spending**

The owner or another authorized user defines the project budget. Recorded expenses reduce the available balance and remain connected to supporting information so project spending can be followed over time.

6. **Communication and Notifications**

Project events can generate notifications for the relevant participants through the web or mobile application. This keeps users informed about assignments, approvals, issues, changes, and other activities.

Development Approach

The project will be developed incrementally so that major parts of the platform can be designed, implemented, integrated, and evaluated in stages. The following phases describe the planned development direction rather than a claim that all phases are complete at the proposal stage.

Phase	Planned Work
Phase 1 – Foundation	Authentication, user accounts, projects, basic tasks, documents, issues, site reports, notifications, and audit logging.
Phase 2 – Scheduling and Coordination	Dependencies, critical path, milestones, design changes, reviews, budget and spending workflows, messaging, and reminders.
Phase 3 – Field and Mobile	Flutter mobile workflows, field evidence, voice capture, push notifications, and Arabic/English support.
Phase 4 – Voice and File Intelligence	Voice workflow, file identification, document classification, and document question answering.
Phase 5 – IFC Intelligence	IFC parsing, model hierarchy, model viewing, revision comparison, impact analysis, and coordination findings.

Phase 6 – AI Tools and Realtime	Governed AI tools, specialized agents, run records, and realtime project updates.
Phase 7 – Configurable Authorization	Roles, disciplines, parties, and flexible project/user permission controls.
Phase 8 – Integration and Hardening	Further integration, reliability improvements, controlled advanced AI analysis, dedicated processing where needed, and preparation of the mobile application for release.

Evaluation Criteria

- Authorization correctness: verify that users receive only the permissions appropriate to their roles and project access.
- Functional coverage: evaluate the major web, backend, and mobile workflows.
- Scheduling correctness: verify dependency and critical-path behavior using known task relationships.
- AI safety and control: verify that AI tools respect authorization and require confirmation before important project changes.
- Voice workflow quality: evaluate transcription and interpretation for Arabic, English, and mixed-language field input.
- Robustness: evaluate document processing, model processing, worker behavior, and realtime communication under expected failure conditions.
- Internationalization: evaluate Arabic and English interfaces and right-to-left behavior.
- Traceability: verify that important project actions and decisions retain appropriate history and supporting evidence.
- Reproducibility: verify that the project can be started using the documented development setup.

What Makes the Project Different

- It connects the main participants in a construction project within one platform instead of focusing on only one stakeholder group.
- Its permission model is configurable, allowing different organizations to adapt roles and responsibilities to their own structure.
- Project information is connected across tasks, schedules, documents, field evidence, design changes, budget activity, approvals, and communication.
- Engineering model changes are treated as project information that can be connected to actual project work.
- Voice and AI assistance can reduce repetitive field and information-management work while keeping users responsible for important decisions.
- AI document assistance follows project permissions rather than creating a separate unrestricted information channel.
- Specialized AI agents focus on bounded project activities such as site reports, document consistency, revision impact, requests, and schedule risk.
- Budget and spending information can give the owner greater visibility into how project funds are being used.

- Important actions and decisions are designed to remain traceable through project history and supporting evidence.
- The platform connects office workflows with field workflows through web and mobile applications.

Expected Outcome

The expected outcome of the project is a unified construction project management and collaboration platform that brings project participants, information, and workflows together in one environment. The platform is intended to make project communication more organized, improve visibility into project activities and spending, connect design and construction information, simplify field reporting, and support faster access to project knowledge.

The project is also expected to demonstrate how AI can be introduced into construction project workflows in a controlled manner. AI assistance will focus on practical tasks such as voice processing, document question answering, and specialized analysis while keeping permissions, human review, and project traceability at the center of the system.

Conclusion

The Smart Construction Platform is proposed as a unified environment for managing construction projects and connecting the people responsible for them. Its central idea is to bring owners, contractors, engineers, consultants, project managers, site teams, and administrators into the same project workflow while allowing each participant to access the information and actions appropriate to their role.

The platform combines project management, scheduling, a central project library, field reporting, communication, notifications, budget and spending visibility, engineering model intelligence, and AI-assisted workflows. By connecting these capabilities, the project aims to reduce fragmented information, improve traceability, make project changes easier to follow, and provide users with a clearer view of what is happening across the project.

The proposal focuses on a practical and controlled use of technology: AI assists users rather than replacing professional judgment, permissions remain under organizational control, and important project activities are intended to remain traceable. The project will be developed and evaluated incrementally according to the proposed objectives and workflows.